

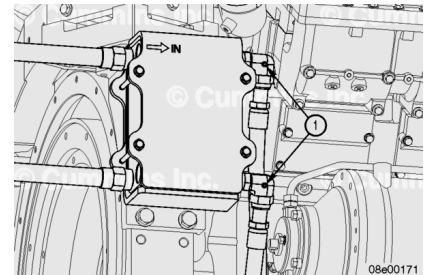
Trainings ▾

Initial Check

Remove plugs (1) in elbows and replace with mechanical pressure gauges.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.



⚠ CAUTION ⚠

Coolant will leak when plug is removed. If necessary, drain coolant before removing plugs. Fill the cooling system before continuing on with this check. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/28/28-008-018-tr.html>)

Operate the engine at rated speed. Record the readings on the mechanical pressure gauges. The difference in the readings should **not** exceed 13 kPa [2 psi].

If the pressure drop is not within specification, the gear cooler must be cleaned or replaced.

Remove the mechanical pressure gauges and install the plugs into the elbows.

Torque Value: 15 n•m [133 in-lb]

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing,

remove the negative (-) battery cable first and attach the negative (-) battery cable last.

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

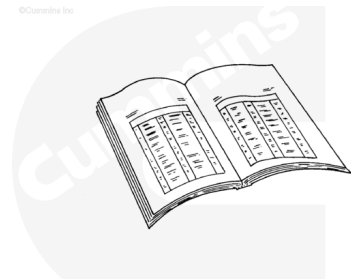
⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

⚠ CAUTION ⚠

Use caution when draining coolant that the coolant is not spilled or drained into the bilge area. Do not pump coolant overboard. If the coolant is not reused, it must be disposed of in accordance with local environmental regulations.

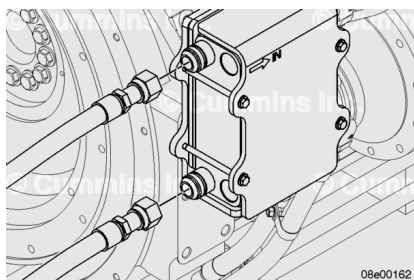
- Disconnect the batteries or air supply line to the air starter to prevent accidental engine starting. Refer to the OEM service manual.
- Drain the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/28/28-008-018-tr.html>)
- Drain the oil from the marine gear oil cooler. Refer to the marine gear manufacturers instructions for draining the oil.
- Disconnect the coolant lines. Refer to Procedure 008-137 in Section 8. (</qs3/pubsys2/xml/en/procedures/28/28-008-137-tr.html>)



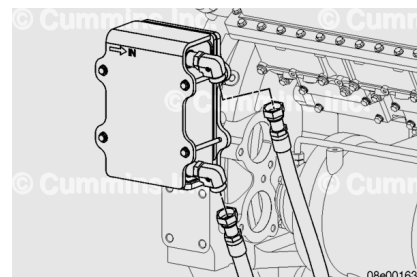
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Remove

Disconnect marine gear oil lines from the marine gear oil cooler.

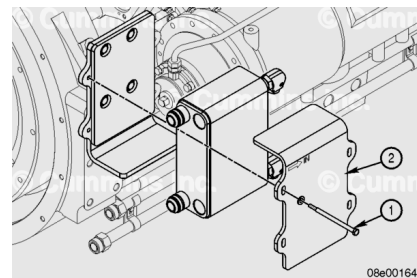


Note : Only remove the elbows from the marine gear oil cooler if necessary.



⚠ CAUTION ⚠

Once the top bracket is removed, the marine gear cooler is unsecured. Remove the marine gear cooler as soon as the top bracket is removed.

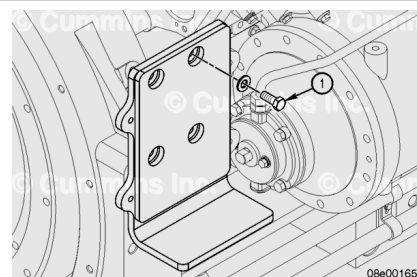


Note : Retain the rubber pad for reuse.

Remove the four cap screws (1), washers, and the top bracket (2) that hold the marine gear cooler to the bottom bracket.

Remove the marine gear oil cooler.

If necessary, remove the four cap screws (1), washers, and bottom bracket.



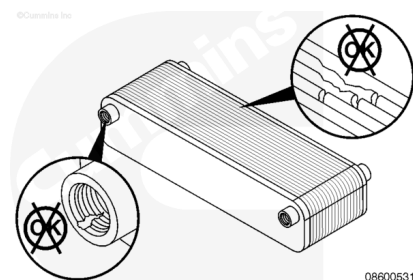
Inspect for Reuse

Inspect the marine gear oil cooler rubber pad for rips, tears, and other damage. If the rubber pad is damaged, it should be replaced.

If the rubber pad is damaged, remove it and inspect the marine gear oil cooler for damage.

Inspect the marine gear oil cooler plates and shell for damage. If the plates or shell is damaged, the marine gear oil cooler **must** be replaced.

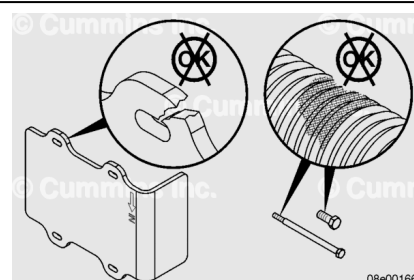
Inspect the marine gear oil cooler coolant and oil ports for thread damage. If the threads are damaged beyond repair, the marine gear oil cooler **must** be replaced.



Inspect the brackets for cracks and other damage.

Inspect the capscrews for damaged threads.

If damage is found, the part **must** be replaced.

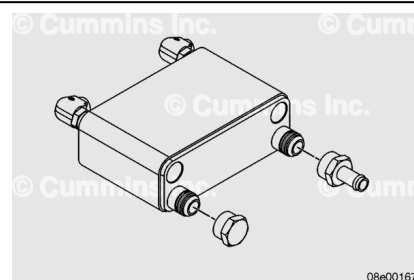


Pressure Test

Install the coolant and oil connectors into the marine gear oil cooler.

Place a JIC 20 (1 5/8-12) threaded cap onto one of the oil connectors of the marine gear oil cooler.

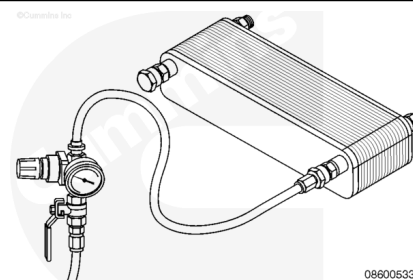
Place a JIC 20 (1 5/8-12) threaded cap with an air fitting onto the remaining oil connector of the marine gear oil cooler.



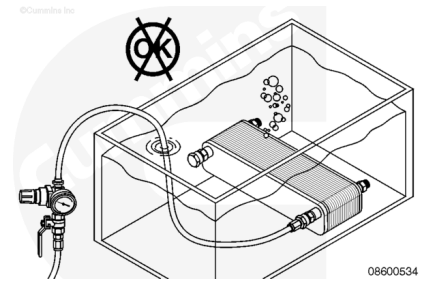
Install a pressure regulator onto the cap and air fitting.

Connect shop air to the pressure regulator.

Adjust the pressure regulator to 170 kPa [25 psi].



To reduce the possibility of damage to the marine gear oilcooler, the water in the test tank must be treated with ananocorrosive additive.



Note : Make sure the marine gear cooler sits flat on the bottom of the tank, as illustrated, so that air bubbles can escape.

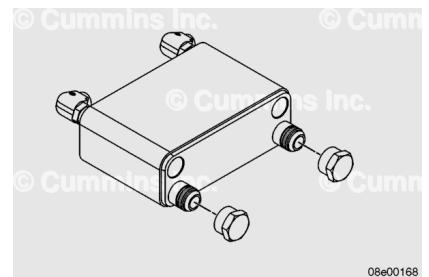
Submerge the marine gear oil cooler in a tank of treated water for 10 minutes.

If air bubbles escape from the marine gear oil cooler in the last two minutes of the test, the marine gear oil cooler **must** be replaced.

Remove the marine gear oil cooler from the tank of treated water.

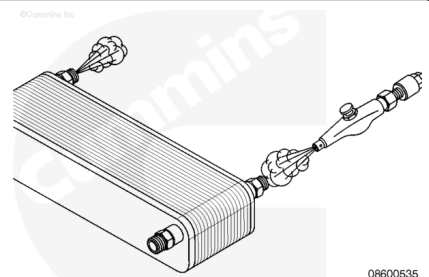
If a leak on the coolant side is suspected, repeat the pressure test, pressurizing the coolant side and placing caps on the oil side.

Note : Make sure to cap the gear oil connections before placing the cooler in tank, so that the oil side does not get contaminated with the treated water.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying dirt and debris can cause personal injury.



Remove the pressure regulator from the air fitting.

Remove the caps from the oil connectors of the marine gear oil cooler.

Drain the treated water out of the marine gear oil cooler.

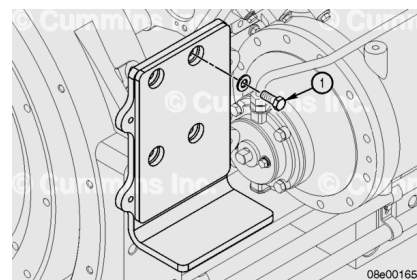
Use compressed air to dry the coolant side of the marine gear oil cooler.

Install

If removed, install the bottom bracket, washers, and capscrews (1).

Tighten the capscrews.

Torque Value: 95 n•m [70 ft-lb]



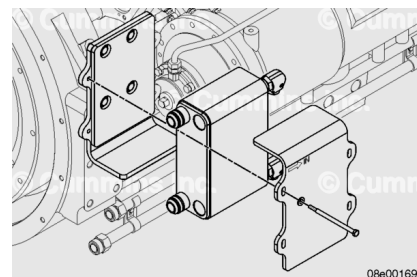
If removed, install the rubber pad around the oil cooler. Place the marine gear oil cooler onto the rear bracket. Make sure that the coolant connections are facing the front of the engine.

Install the front bracket, washers, and capscrews.

Hand-tighten the capscrews and make sure that the bracket is squarely contacting the gear cooler.

Tighten the capscrews.

Torque Value: 38 n•m [28 ft-lb]

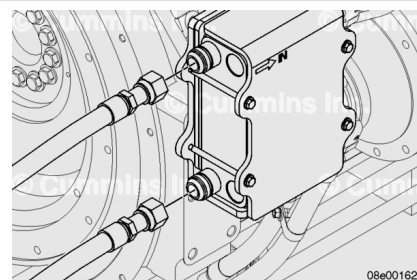


Note : Make sure the oil flow is into the top fitting on the marine gear oil cooler.

Connect the gear oil lines to the marine gear oil cooler connectors.

Tighten the gear oil lines.

Torque Value: 74 n•m [55 ft-lb]



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



- Connect the coolant lines. Refer to Procedure 008-137 in Section 8. (</qs3/pubsys2/xml/en/procedures/28/28-008-137-tr.html>)

- Connect the batteries or air starter air supply line. Refer to the OEM service manual.
- Fill the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/28/28-008-018-tr.html>)
- Add gear oil to the marine gear. Refer to the OEM service manual.
- Operate the engine and check for leaks.

Last Modified: 23-Nov-2010
